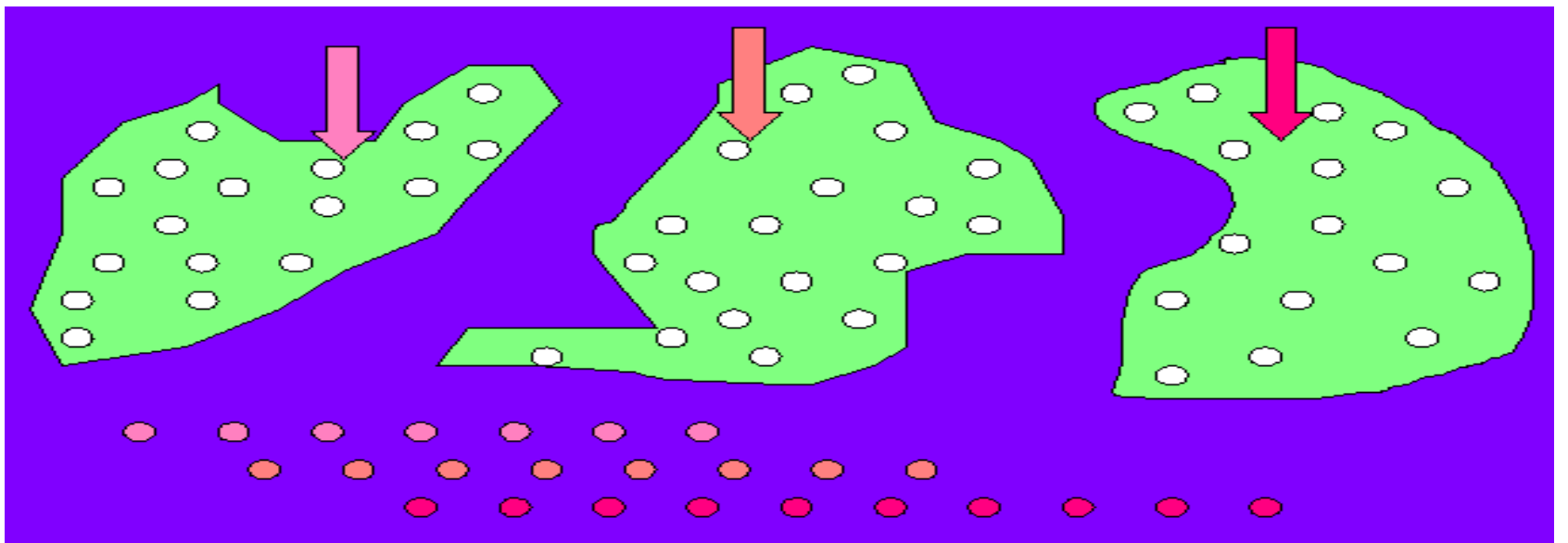
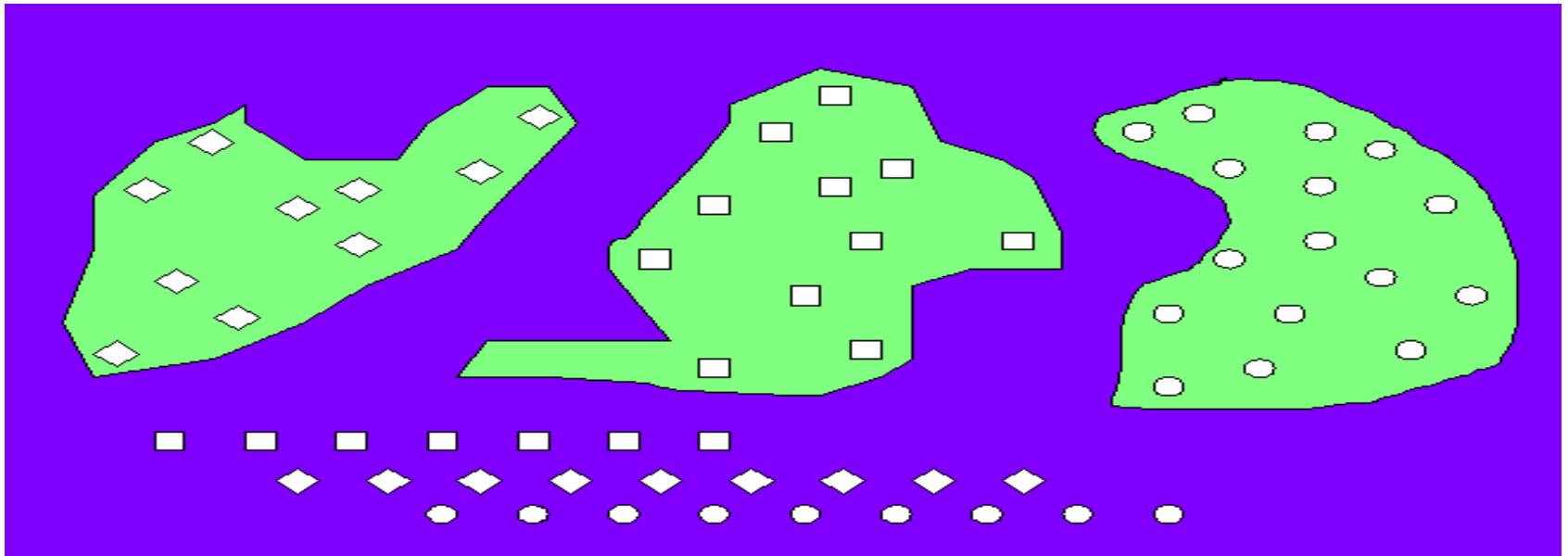


Model of multi-independent samples



Hypothesis tests for several independent samples

- Compare several proportions
- Compare mean values of several populations

Compare several proportions

Let X be a binary variable taking two values 0 and 1 .
Collecting data from that variable under k different conditions we have a sample containing k groups of observations related with the conditions

Let $p_1, p_2, \dots, p_k$

be probabilities of appearance of value 1 of variable X under each of the above k conditions.

Hypothesis $H : p_1 = p_2 = \dots = p_k$

Alternative Hypothesis

K: there is certain difference between $p_1, p_2, \dots, p_k$

Data: Perform a $2 \times k$ table of 2 rows and k columns: each column for one group, the 1st row for value 1, the 2nd row for value 0 of the variable at observations:

Table 1. Observed frequency

	Group 1	Group2		Group k	
X = 1	n_{11}	n_{12}	. . .	n_{1k}	n_1
X = 0	n_{01}	n_{02}	. . .	n_{0k}	n_0
	$n^{(1)}$	$n^{(2)}$		$n^{(k)}$	n

$$n_1 = n_{11} + n_{12} + \dots + n_{1k} \quad ; \quad n_0 = n_{01} + n_{02} + \dots + n_{0k}$$

$$n^{(j)} = n_{j1} + n_{j0} \quad ; \quad j = 1, 2, \dots, k \quad ; \quad n = n_0 + n_1$$

Compare several proportions

- If the hypothesis is correct, the proportion of occurrence of **1** estimated commonly to all columns (conditions) is equal to

$$n_1 / n = p^{\wedge}$$

- The proportion of occurrence of **0** estimated commonly to all columns is equal to

$$n_0 / n = 1 - p^{\wedge}$$

Perform the table of expected (theoretical) frequencies of the hypothesis:

Table 2. predicted (expected) frequency

	Group 1	Group2		Group k	
$X = 1$	$n^{(1)} * \frac{n_1}{n}$	$n^{(2)} * \frac{n_1}{n}$	$\dots$	$n^{(k)} * \frac{n_1}{n}$	n_1
$X = 0$	$n^{(1)} * \frac{n_0}{n}$	$n^{(2)} * \frac{n_0}{n}$	$\dots$	$n^{(k)} * \frac{n_0}{n}$	n_0
	$n^{(1)}$	$n^{(2)}$		$n^{(k)}$	n

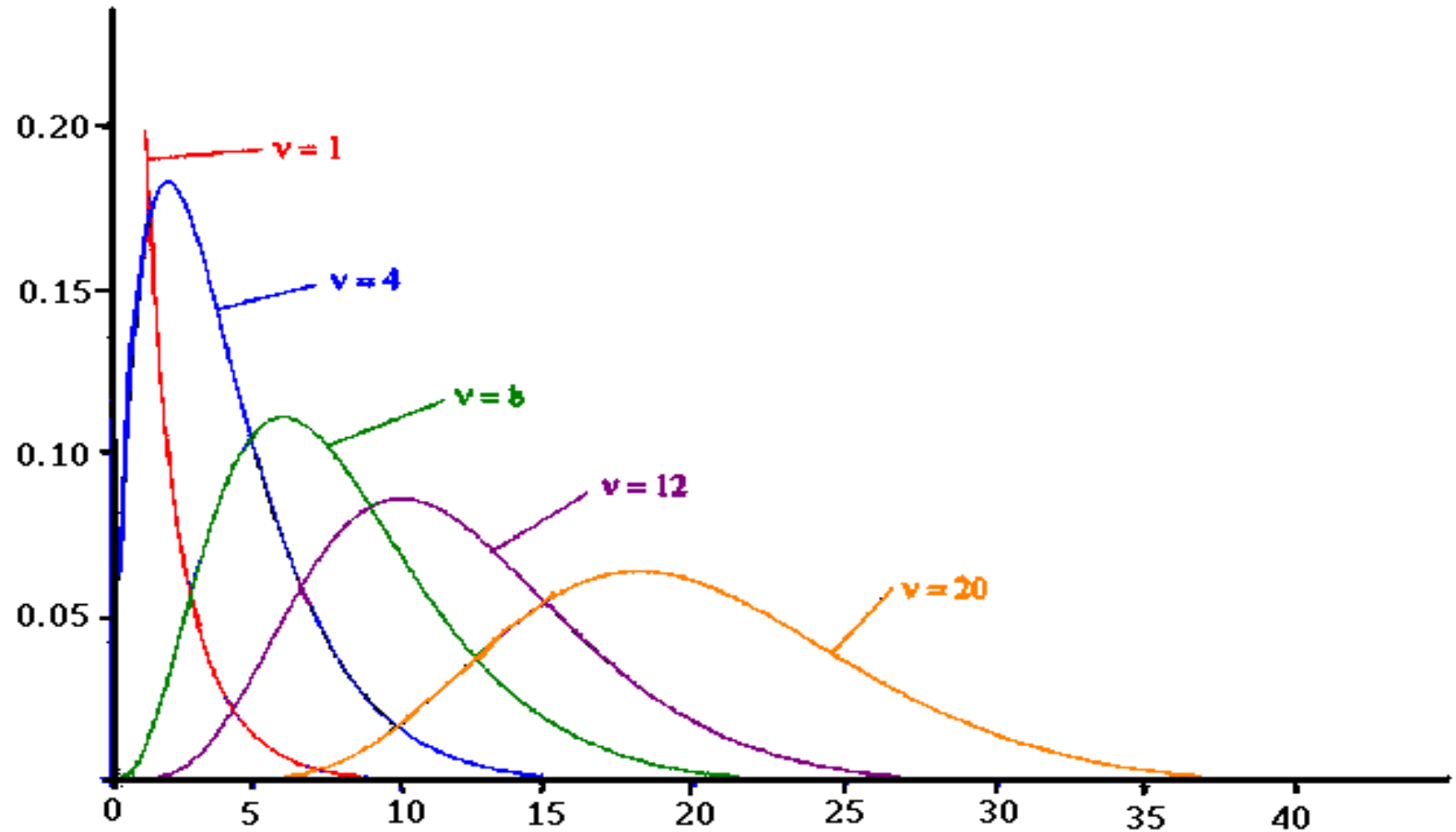
Perform the table of the test statistic:

	Group 1	Group2		Group k
X = 1	$\frac{\left(n_{11} - \frac{n^{(1)}n_1}{n}\right)^2}{\frac{n^{(1)}n_1}{n}}$	$\frac{\left(n_{12} - \frac{n^{(2)}n_1}{n}\right)^2}{\frac{n^{(2)}n_1}{n}}$	...	$\frac{\left(n_{1k} - \frac{n^{(k)}n_1}{n}\right)^2}{\frac{n^{(k)}n_1}{n}}$
X = 0	$\frac{\left(n_{01} - \frac{n^{(1)}n_0}{n}\right)^2}{\frac{n^{(1)}n_0}{n}}$	$\frac{\left(n_{02} - \frac{n^{(2)}n_0}{n}\right)^2}{\frac{n^{(2)}n_0}{n}}$	...	$\frac{\left(n_{0k} - \frac{n^{(k)}n_0}{n}\right)^2}{\frac{n^{(k)}n_0}{n}}$

$$\chi^2 = \sum_{j=1}^k \sum_{i=0}^1 \left(n_{ij} - \frac{n_i \cdot n^{(j)}}{n} \right)^2 / \left(\frac{n_i \cdot n^{(j)}}{n} \right)$$

LEMMA. Suppose that hypothesis H is true.
Then variable χ^2 has distribution approximate
to the Chi-square distribution with $(k - 1)$
degrees of freedom $\chi^2_{(k-1)}$.

Density function of Chi – squared distribution



Using Excel to Compute *Chi – squared* Distribution

- Excel has two functions for computing cumulative probabilities and x values for any *Chi - squared* distribution:
 - CHI.DIST is used to compute the cumulative probability given an x value, p-value.
 - CHI.INV is used to compute the x value given a cumulative probability, critical value.

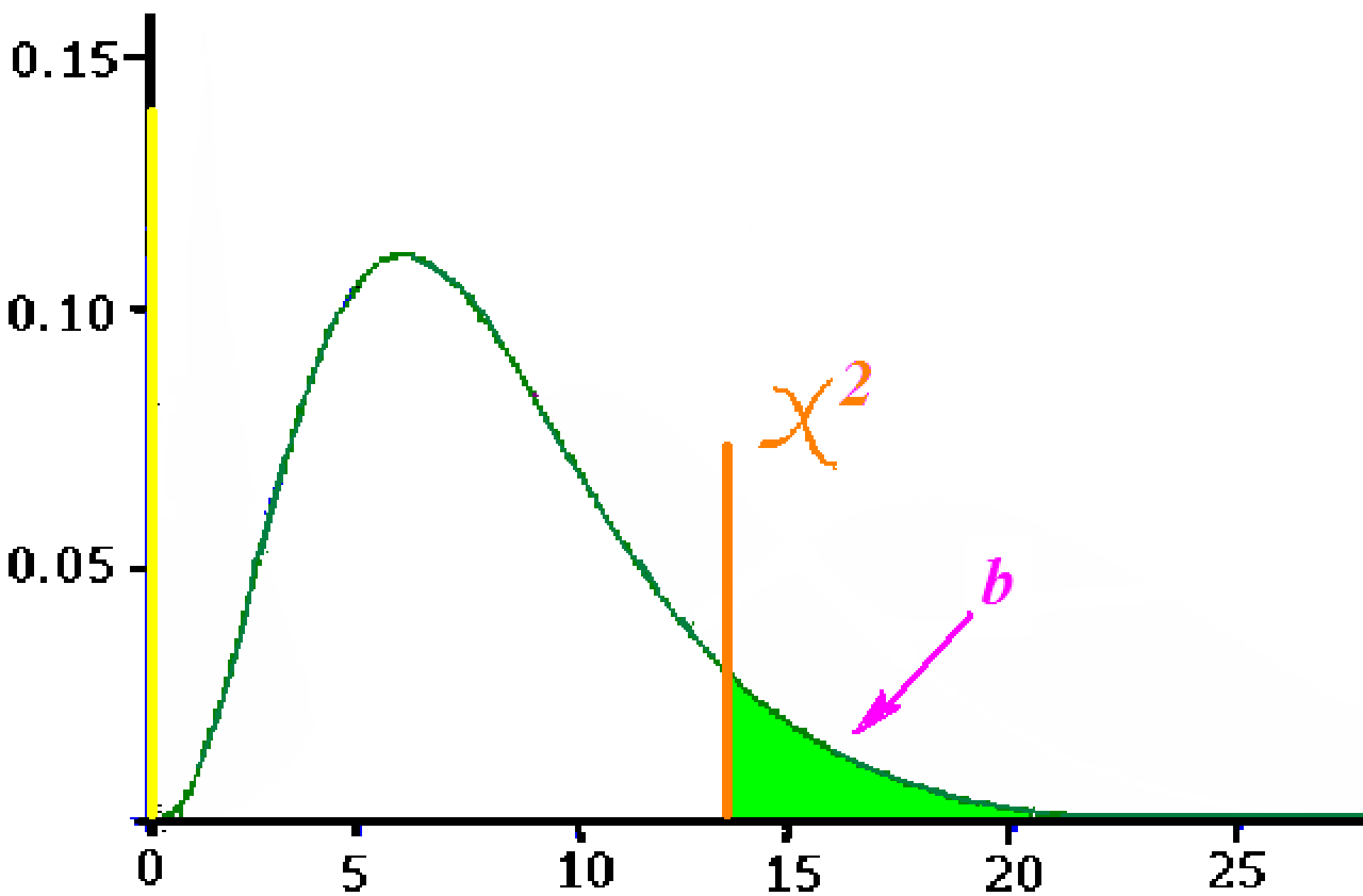
Method A (p-value):

Step 1. Taking a variable $\chi^2_{(k-1)}$ of Chi-squared distribution with $(k-1)$ degrees of freedom calculate the probability (p-value)

$$b = P \{ \chi^2_{(k-1)} > \chi^2 \}.$$

Step 2. Compare the probability b to the given ahead significance level α :

- * If $b \geq \alpha \rightarrow$ accept hypothesis H , conclude the all proportions are equal
- * If $b < \alpha \rightarrow$ reject hypothesis H , confirm the appearance of some difference between proportions.



Method B. (Critical value)

Looking in Table of Chi-squared distribution to find critical value $\chi^2_{(k-1)}(\alpha)$ of Chi-squared distribution with $k-1$ degrees of freedom (α is a given ahead significance level = 5%, 1% or 0.5%)

Decide

- Reject Hypothesis H_0 = if
$$\chi^2 \geq \chi^2_{(k-1)}(\alpha)$$
- Accept Hypothesis H_0 = if
$$\chi^2 < \chi^2_{(k-1)}(\alpha)$$

One-way Analysis of Variance

The one-factor analysis of variance (ANOVA) test is an extension of the T-test method that compares the mean values to the **multi-independent samples' model**: For normally distributed random variables $X^{(1)}, X^{(2)}, \dots, X^{(k)}$ with common variance and expectations $\mu_1, \mu_2, \dots, \mu_k$, consider the hypothesis:

$$H : \mu_1 = \mu_2 = \dots = \mu_k$$

Analysis of Variance:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

H_a : Not all population means are equal

- If H_0 is rejected, we cannot conclude that *all* population means are equal.
- Rejecting H_0 means that at least two population means have different values.

Analysis of Variance:

- Assumptions for Analysis of Variance
 - For each population, the response (dependent) variable is normally distributed.
 - The variance of the response variable, denoted σ^2 , is the same for all of the populations.
 - The observations must be independent.

One-way Analysis of Variance

Consider the sample $\{x_j^{(i)}, j = 1, 2, \dots, n_i\}$ of observations in the i -th group, corresponding to the variable $X^{(i)}$, including n_i observation, $i = 1, 2, \dots, k$.

Consider the model
$$x_j^{(i)} = \mu_i + \varepsilon_{ij} = \mu + \alpha_i + \varepsilon_{ij},$$

where ε_{ij} is the deviation of each observation in the group from the group mean μ_i , and μ the common mean of all observations in the data, α_i is the difference between the group mean μ_i and the overall mean, μ obviously

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = 0$$

One-way Analysis of Variance

The total volatility in the i -th group, $i = 1, 2, \dots, k$, is equal to

$$SS_i = \varepsilon_{i1}^2 + \varepsilon_{i2}^2 \dots + \varepsilon_{in_i}^2$$

and

$$(\varepsilon_{i1}^2 + \varepsilon_{i2}^2 \dots + \varepsilon_{in_i}^2) / (n_i - 1)$$

is the adjusted variance of the random variable $X^{(i)}$.

$$RSS = SS_1 + \dots + SS_k = (\varepsilon_{11}^2 + \varepsilon_{12}^2 \dots + \varepsilon_{1n_1}^2) + \dots + (\varepsilon_{k1}^2 + \varepsilon_{k2}^2 \dots + \varepsilon_{kn_k}^2)$$

is the sum of all volatility in all groups.

Denote the total number of observations by $n = n_1 + \dots + n_k$

We can take

$$RSS / (n - k)$$

as an estimate of the common variance of all variables

$$X^{(1)}, X^{(2)}, \dots, X^{(k)}$$

One-way Analysis of Variance

$$BSS = n_1\alpha_1^2 + n_2\alpha_2^2 + \dots + n_k\alpha_k^2$$

is the total variation between the group means

$$BSS / (k - 1)$$

reflecting the random variation from group to group of the considered quantity .

Ratio between two variances

$$F = \frac{BSS / (k - 1)}{RSS / (n - k)}$$

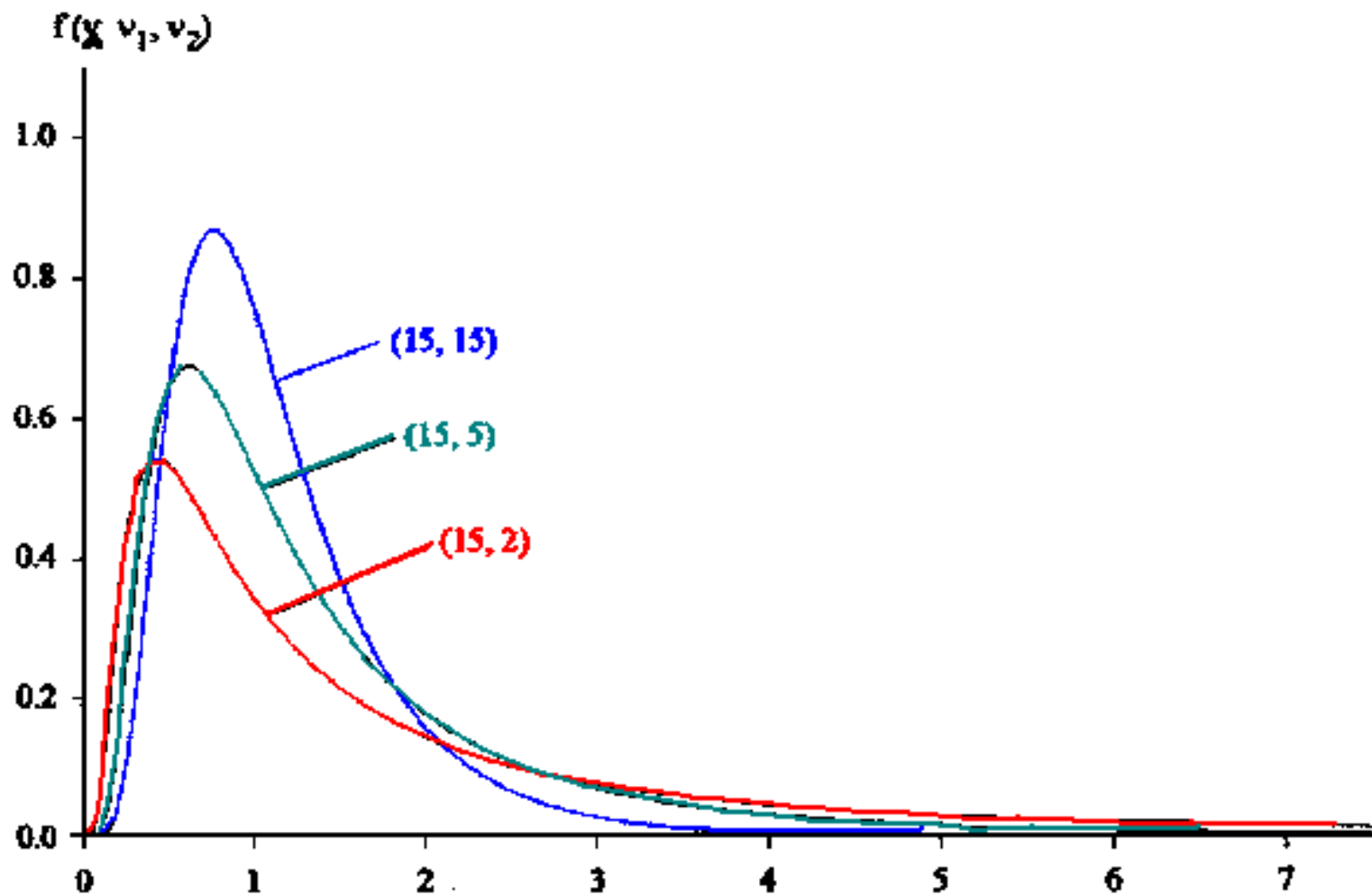
is the test statistic for hypothesis H comparing mean values.

One-way Analysis of Variance

Lemma. *If the **hypothesis H is true**, then the ratio (test statistic) **F** has a Fisher-Snedecor distribution with $(k-1)$ and $(n-k)$ degrees of freedom.*

With the above lemma, the testing problem has approaches of p-value or critical value.

Density function of Fisher-Snedecor distribution (F)

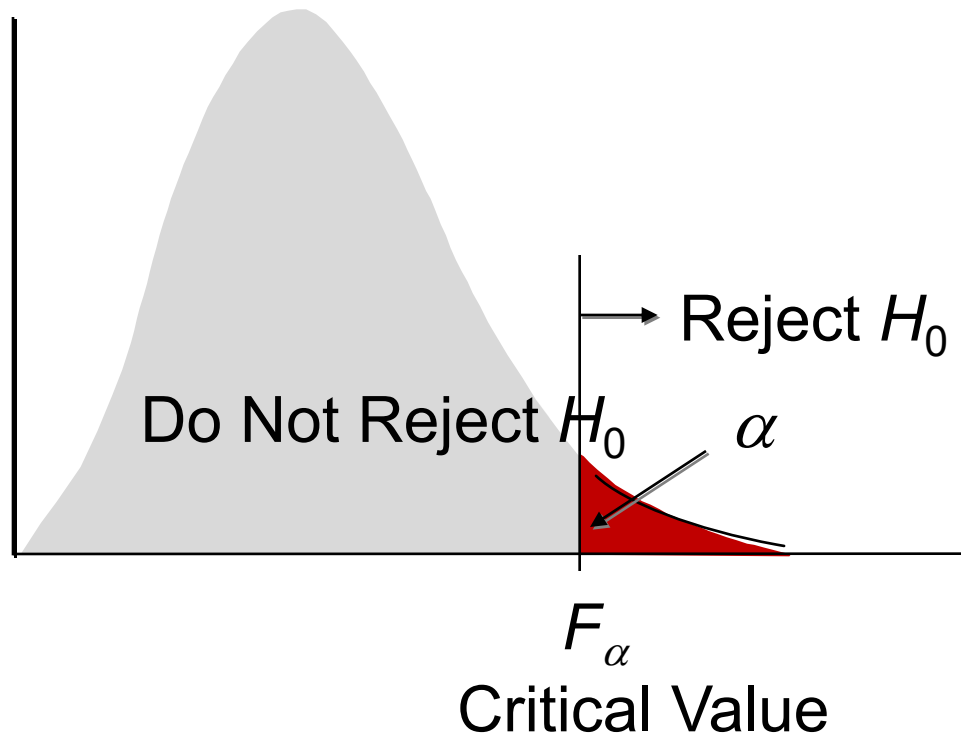


One-way Analysis of Variance

a) Compare the value of the test statistic F with the critical value $F_{cr} = F_{\alpha}^{(k-1, n-k)}$, which is the $(1 - \alpha)$ percentile of the Fisher-Snedecor distribution with $(k-1)$ and $(n-k)$ degrees of freedom: -

- If $F \geq F_{cr} \rightarrow$ rejects hypothesis H ,
- If $F < F_{cr} \rightarrow$ accepts H .

The F Test



One-way Analysis of Variance

b) With ***FS*** being a random variable with a Fisher-Snedecor distribution with $(k-1)$ and $(n-k)$ degrees of freedom , calculate the probability of significance

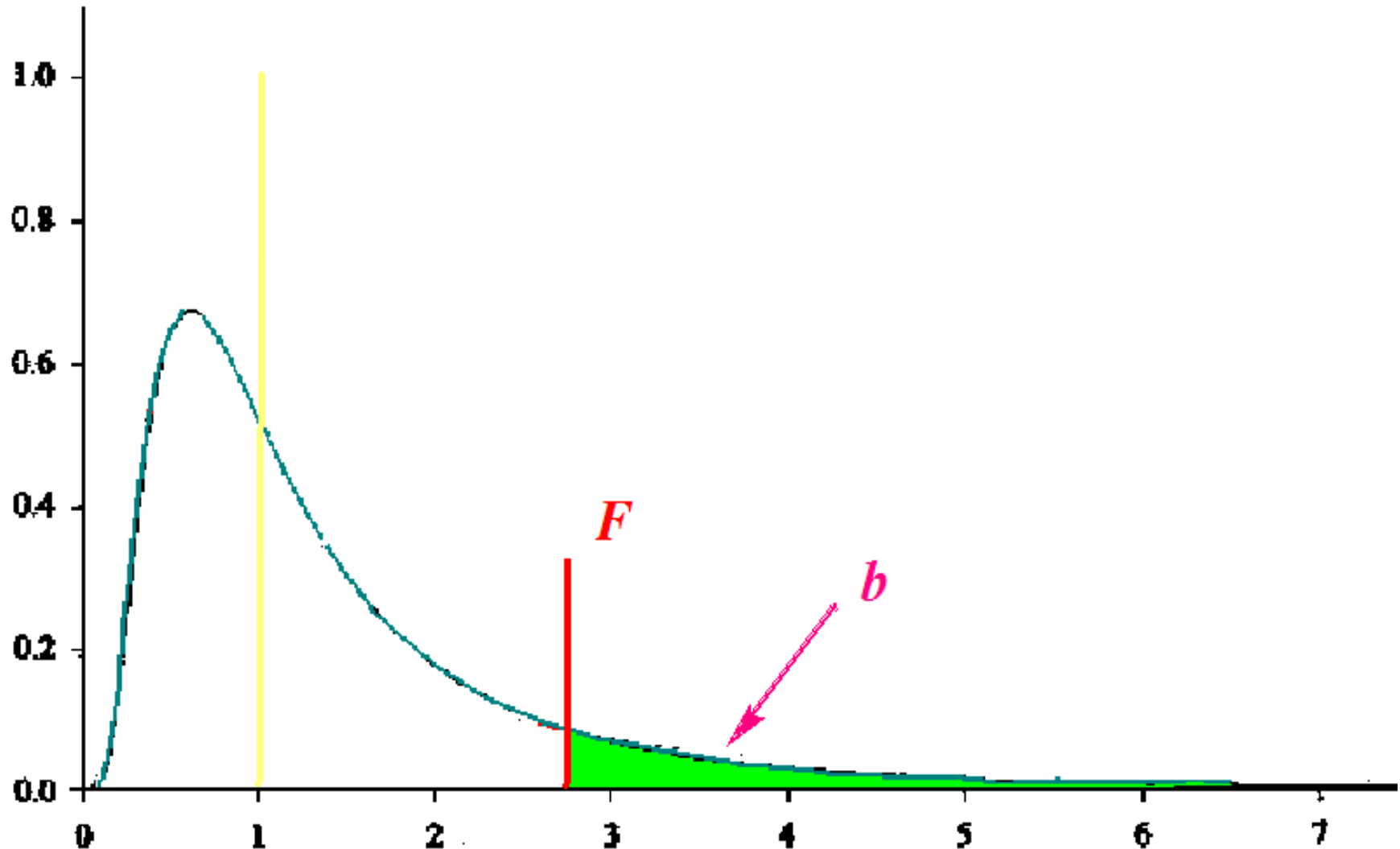
$$p_{sig} = P\{FS > F\}$$

Compare the probability of significance of the test statistic with the significance level α (=5%):

-If $p_{sig} \leq \alpha$ $\rightarrow$ rejects hypothesis H_0 ,-

- If $p_{sig} > \alpha$ $\rightarrow$ accepts H_0 .

The right-tailed probability of Fisher-Snedecor distribution (F)



Using Excel to Compute *Fisher - Snedecor* Distribution

- Excel has two functions for computing cumulative probabilities and x values for any *Fisher - Snedecor* distribution:
 - F.DIST is used to compute the cumulative probability given an x value, p-value.
 - F.INV is used to compute the x value given a cumulative probability, critical value.